# Mritra

Catching cross-bank money-mule chains without banks exposing their data

IBM Z Datathon 2026 | Theme B: AI Secured: Innovation Without Exposure | Team lead: Rehan, Sanjay Ghodawat University, Kolhapur

## 1. The problem

When someone is cheated online, the stolen money rarely stays put. Within hours it is moved through chains of ordinary-looking accounts spread across many banks, then withdrawn. Each bank sees only its own small step, and that step looks normal. The whole pattern is visible only to someone who can see every bank at once, and no one is allowed to: customer data is private, and banks cannot safely hand it to each other or to a central party.

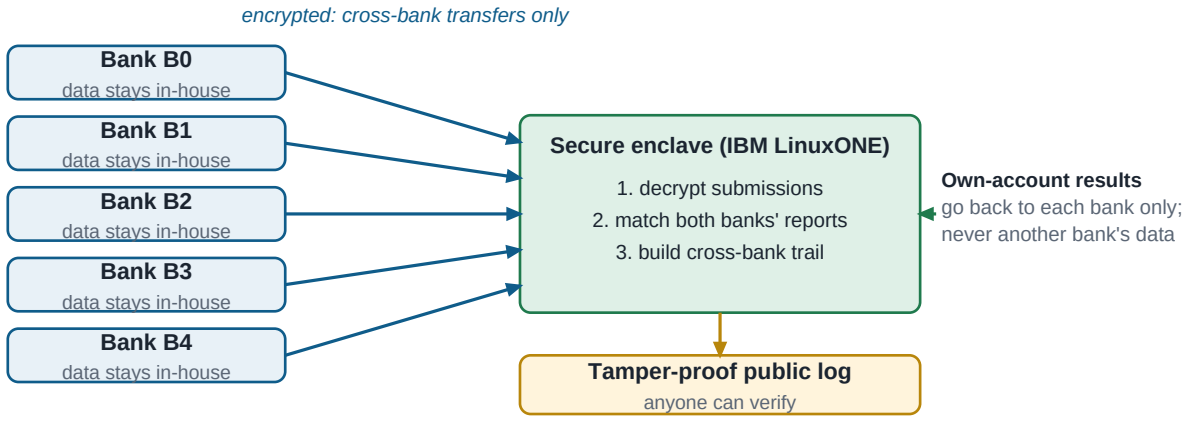
- The Financial Action Task Force's 2024 assessment of India identifies cyber-enabled fraud as its top money-laundering risk.
- Research from the Royal United Services Institute (RUSI) indicates about half of mule-account money leaves the receiving account within an hour, so detection has to be fast.
- Investigations by the CBI have found bank insiders complicit, so a design that relies on trusting one party is not enough.

## 2. Why existing approaches fall short

- **Each bank alone** cannot see transfers that leave its books.
- **Federated learning** lets banks train a shared model, but it shares what patterns look like, not the missing transactions. A bank still scores only its own fragment.
- **A central database** of all transactions solves visibility but is unacceptable for privacy, competition and insider-misuse reasons.

## 3. The solution

Banks keep their data. Each hour, each bank sends a protected package containing only the transfers that cross to other banks, to a secure computing environment (an enclave) on IBM LinuxONE. The enclave matches the two sides of every transfer, which exposes any bank that hides or alters one, connects the pieces into the full trail, and sends each bank back results about its own accounts only. Every step is written to a tamper-proof log that anyone can verify.




What each bank gets: a risk signal for its own accounts, such as the account's role in a chain (collecting, passing along, cashing out), the suspected path, the amount at risk, and the time left before the money exits.

## 4. How we prove it works

We run four setups with the same model. They differ only in what the model is allowed to see, so any difference comes from visibility, not tuning.

| Setup | What it sees | Role |
|---|---|---|
| C1 Siloed | Own bank's data only | Today's reality |
| C2 Federated | Own data; model trained jointly | Does shared training alone close the gap? |
| **C3 Mritra** | Own data plus enclave's cross-bank signals | Our solution |
| C4 Oracle | The entire network | Upper bound |

**Headline measure:** the share of the gap between Siloed and Oracle that Mritra recovers on cross-bank laundering. We commit in advance that success means this share is above zero with a confidence interval that excludes zero under two different ways of splitting the data across banks. If shared training (C2) matches Mritra, or the gap does not exist, we report that honestly.

We also report how early cases are flagged before money exits, how many innocent customers are flagged, how often a deliberately dishonest bank is caught, and how much data each bank discloses per round.

## 5. Data and honesty statement

- **Data:** IBM AMLworld (HI-Small), a public synthetic dataset with labelled laundering patterns. The banks in it are grouped into five simulated institutions. Data is split by time (60/20/20), never at random.
- **No real customer or bank data is used.**
- If hardware-backed secure execution is unavailable on the provided system, the enclave runs as a clearly labelled simulation. We will not claim protections we do not have.

## 6. Who it is for

Bank fraud and AML teams (users); bank risk and compliance heads (buyers); RBI and FIU-IND (regulators); NPCI and I4C (shared infrastructure and fraud response); law-enforcement cyber cells (investigators); auditors and privacy bodies (trust).

## 7. Plan to the event

- By 8 Oct: confirm enough laundering crosses banks, and that seeing the whole network helps (the Oracle beats Siloed).
- 9-16 Oct: build bank services, enclave, log and dashboard; set up LinuxONE.
- 17-18 Oct: run final experiments, demo live, present results.